

System 10 – Solar Energy Generation and Operations Management

Project Scenario

You are engaged by **SolarSphere Energy**, a multinational renewable energy company operating and managing large-scale solar photovoltaic (PV) power plants across North America, Europe, the Middle East, Africa, and Asia-Pacific.

SolarSphere owns and manages:

- Utility-scale solar farms
- Distributed solar generation facilities
- Battery Energy Storage Systems (BESS)
- Grid interconnection infrastructure
- Regional Operations Control Centers
- Asset maintenance and field service operations

The company generates electricity for utility companies, commercial customers, energy markets, and power purchase agreement (PPA) clients. SolarSphere requires a unified enterprise architecture to support:

- Real-time monitoring of solar generation assets and performance
 - Solar plant operations and energy production optimization
 - Grid integration and energy dispatch management
 - Predictive maintenance and asset lifecycle management
 - Weather forecasting and generation forecasting
 - Battery energy storage monitoring and optimization
 - Environmental, sustainability, and ESG reporting
 - Integration with ERP, SCADA, Asset Management, Energy Trading, and Grid Management systems
-

Project Objectives

1. Develop a **layered reference architecture** covering Information Architecture and Application Architecture for SolarSphere's renewable energy operations ecosystem.
2. Ensure traceability from business goals including:
 - Energy generation efficiency
 - Asset availability
 - Grid reliability
 - Revenue optimization
 - Sustainability performance

- Regulatory compliance

through data models, applications, services, and integration flows.

3.

Produce professional diagrams, API definitions, and documentation suitable for an architecture review board in the renewable energy sector.

Team Roles & Responsibilities

Each member should own one or more of these roles:

Enterprise Architect

- Define architectural vision and guiding principles for utility-scale solar operations.
- Establish interactions between operational technology (OT) and enterprise IT systems.
- Align digital transformation initiatives with renewable energy business objectives.

Information Architect

- Model solar generation, weather, grid, maintenance, battery storage, sustainability, and financial data domains.
- Define Master Data Management (MDM) strategies for assets, sites, customers, and grid connections.
- Specify metadata management and data governance practices.

Application Architect

- Design business capabilities and application service boundaries.
- Define operational, analytical, forecasting, maintenance, and reporting applications.
- Create API specifications for operational and analytical services.

Integration Architect

- Design integrations between SCADA, EMS, ERP, EAM, forecasting systems, energy market platforms, and cloud analytics.
- Recommend event-driven, streaming, and batch integration patterns.
- Define real-time telemetry ingestion architectures.

Quality & Non-Functional Lead

- Define cybersecurity, availability, scalability, resilience, and compliance requirements.

- Establish operational SLAs and monitoring strategies.
 - Design observability and disaster recovery architectures.
-

Documentation & Presentation Lead

- Consolidate architecture deliverables.
 - Ensure consistency and traceability across artifacts.
 - Prepare executive presentations and reporting materials.
-

Deliverables & Artifacts

All diagrams must use a standard notation (UML, C4, or ArchiMate).

1. Architecture Vision (2 Pages)

Solar Energy Context

Describe:

- Global renewable energy operations
- Distributed energy generation
- Utility-scale solar farms
- Battery storage integration
- Smart-grid connectivity

Business Drivers

- Maximize energy generation
- Improve plant performance ratio
- Increase asset availability
- Optimize battery utilization
- Improve energy forecasting accuracy
- Reduce maintenance costs
- Achieve ESG and sustainability targets

Architectural Principles

Examples:

- Renewable Operations First
- Cloud-Enabled Analytics

- Data-Driven Decision Making
- Secure OT/IT Integration
- Sustainability by Design
- Real-Time Visibility

Include

- Scope
 - Stakeholders
 - Business capability model
 - High-level roadmap
-

2. Information Architecture Package

Conceptual Data Model

Asset Domain

- Solar Farm
- Solar Array
- String Inverter
- Central Inverter
- Transformer
- Substation
- Battery Storage Unit
- Weather Station

Operations Domain

- Energy Production
- Generation Event
- Curtailment Event
- Grid Dispatch Event
- Outage Event

Maintenance Domain

- Asset
- Inspection
- Work Order
- Maintenance Activity
- Failure Incident

Forecasting Domain

- Weather Forecast
- Irradiance Forecast
- Generation Forecast
- Market Forecast

Sustainability Domain

- Carbon Offset
- ESG Metric
- Compliance Report
- Environmental Assessment

Commercial Domain

- Customer
- PPA Contract
- Utility Partner
- Market Transaction

Logical Data Model

Include:

- Primary keys
- Foreign keys
- Business rules
- Data ownership
- Normalization assumptions

Physical Data Model Sketch

Design:

Operational Data Stores

- Plant Operations Database
- Asset Management Database
- Forecasting Database
- Commercial Operations Database

Time-Series Data Stores

Store:

- Power output
- Solar irradiance
- Temperature
- Voltage
- Current
- Battery State of Charge (SOC)
- Grid frequency

Include:

- Partitioning strategy
- Indexing strategy
- Retention policies

Master Data Management Strategy

Develop MDM for:

- Solar assets
- Generation sites
- Grid connections
- Customers
- Utility operators
- Equipment vendors

Include:

- Golden records
- Stewardship roles
- Quality controls

Sample Metadata Catalog

Examples:

- Data lineage for SCADA telemetry
- Asset master definitions
- ESG reporting definitions
- Energy forecasting lineage

Business glossary examples:

- Capacity Factor

- Performance Ratio
 - Availability
 - Curtailment
 - State of Charge
 - Levelized Cost of Energy (LCOE)
-

3. Application Architecture Package

Component Diagram

Include:

Operational Systems

- SCADA Platform
- Energy Management System (EMS)
- Plant Monitoring System
- Battery Management System (BMS)

Enterprise Systems

- Enterprise Asset Management (EAM)
- ERP
- Energy Trading Platform
- ESG Reporting Platform

Analytics Systems

- Data Lake
- Forecasting Platform
- AI Optimization Engine
- Performance Analytics Dashboard

User Interfaces

- Operator Dashboard
 - Mobile Maintenance Application
 - Executive Analytics Portal
 - Sustainability Dashboard
-

Service Boundaries

Map bounded contexts to:

- Generation Operations
 - Energy Forecasting
 - Asset Maintenance
 - Battery Optimization
 - Grid Integration
 - Commercial Operations
 - Sustainability Reporting
-

API Contracts

API 1 – Asset Maintenance Service

Functions:

- Create work order
- Schedule maintenance
- Assign technicians
- Close maintenance tasks

API 2 – Generation Monitoring Service

Functions:

- Retrieve generation statistics
 - Asset status queries
 - Site performance metrics
 - Outage notifications
-

Streaming Interfaces

Define MQTT, Kafka, or Event Hub topics for:

- Solar generation telemetry
 - Inverter performance
 - Weather data
 - Battery telemetry
 - Grid dispatch events
 - Outage alerts
-

Operator Workflow

Create a sequence diagram showing:

Solar Irradiance Forecast → Energy Generation → Battery Storage Optimization → Grid Dispatch → Revenue Settlement

Include:

- Forecasting systems
- Operational systems
- Market systems
- Battery management systems

4. Integration & Middleware Architecture

Integration Patterns Matrix

Pattern	Example
Streaming	SCADA → Analytics
CDC	ERP → Reporting Platform
Batch ETL	Operations → Data Lake
Event Driven	Asset Failure Alerts
API Integration	Grid Operator Interfaces
Data Replication	Historian → Cloud Analytics

Workflow Diagram

Model:

Predictive Maintenance Process

Sensor Data → Condition Monitoring → AI Analytics → Failure Prediction → Work Order Generation → Field Maintenance → Asset Return to Service

Forecasting Process

Weather Data → Forecast Models → Generation Forecast → Grid Dispatch Planning → Market Participation

Middleware Rationale

Evaluate:

OT Connectivity

- OPC UA
- MQTT
- IEC 61850 Integration

Enterprise Connectivity

- API Gateway
- ESB
- Event Streaming Platform

Justify technology choices based on scalability, latency, reliability, and operational complexity.

5. Non-Functional Requirements & Quality Attributes

Security Model

Operational Security

- SCADA network segmentation
- Industrial DMZ
- Device authentication
- Secure telemetry transport

Enterprise Security

- Single Sign-On (SSO)
 - Multi-Factor Authentication
 - Role-Based Access Control
 - Encryption in transit and at rest
-

Resilience Tactics

Include:

- Active-active cloud architecture
- Circuit breakers
- Retry policies
- Offline operational capability

- Disaster recovery
-

Scalability Strategy

Address:

- Growth in telemetry volume
 - Expansion to new solar sites
 - Additional battery storage assets
 - AI and forecasting workloads
-

Observability Plan

Include:

- Centralized logging (ELK)
 - Metrics collection (Prometheus)
 - Grafana dashboards
 - Distributed tracing
 - Site-level operational monitoring
-

SLA Definitions & Capacity Planning

Examples:

Metric	Target
SCADA Availability	99.99%
Telemetry Latency	< 2 seconds
Forecast Availability	99.95%
Work Order Response Time	< 5 seconds
Grid Dispatch Notification	< 30 seconds

6. Deployment Topology Diagram

Edge Layer

Include:

- Solar inverters

- Sensors
 - Weather stations
 - Battery controllers
 - SCADA gateways
-

Regional Operations Centre

Include:

- EMS
 - Operational dashboards
 - Grid dispatch coordination
 - Regional monitoring systems
-

Enterprise Cloud

Include:

- API Gateway
 - Data Lake
 - AI/ML Platform
 - Forecasting Services
 - ESG Reporting Platform
-

Compute Platforms

- Kubernetes for microservices
 - Time-series databases
 - Cloud data lake
 - AI/ML platform
 - Stream processing infrastructure
-

Network Architecture

Operational Technology Zone

- Solar plant control networks
- Local SCADA systems

Industrial DMZ

- Secure OT/IT integration

Enterprise IT Zone

- ERP
- Asset Management
- Market Systems

Cloud Analytics Zone

- AI optimization
- Forecasting
- Sustainability reporting

Connectivity

- VPN
 - SD-WAN
 - Cellular backup
 - Multi-region failover
-

7. Executive Presentation (10 Slides)

1. Business Context
 2. Renewable Energy Strategy
 3. Architecture Vision
 4. Business Capability Model
 5. Information Architecture
 6. Application Architecture
 7. Integration Architecture
 8. Security & Resilience
 9. Deployment Architecture
 10. Roadmap & Recommendations
-

8. Reflective Report (1,500 Words)

Discuss:

- Architectural decisions
- Renewable energy operational challenges
- Cloud versus edge design decisions

- Grid integration considerations
 - Forecasting and analytics requirements
 - Cybersecurity and compliance considerations
 - Team collaboration experiences
 - Lessons learned
-

Methodology & Tools

Modeling

- Archi
- Lucidchart
- draw.io

API Design

- Swagger/OpenAPI Editor
- GraphQL Playground
- AsyncAPI Studio
- MQTT Explorer

Data Modeling

- dbdiagram.io
- SQL DDL snippets

Documentation

- Microsoft Word
- PowerPoint (slides and poster)

Version Control & Collaboration

- Git Repository
 - GitHub or Azure DevOps Issues
-

Architectural Focus

Students should demonstrate clear traceability from solar energy business objectives through information architecture, application services, integration architecture, and deployment topology.

Architectural decisions should be justified in the context of:

- Utility-scale renewable energy generation
- Grid reliability and energy dispatch
- Weather and generation forecasting
- Battery storage optimization
- Asset reliability and predictive maintenance
- Environmental sustainability and ESG reporting
- Global operations management
- OT/IT convergence

The final architecture should illustrate how SolarSphere Energy can operate a globally distributed portfolio of solar generation assets while supporting future capabilities such as AI-driven energy optimization, digital twins, autonomous maintenance planning, advanced forecasting, and net-zero sustainability initiatives.